

Overfitting and Underfitting

1. Underfitting

Definition:

Underfitting occurs when a model is too simple to capture the underlying patterns in the data.

Key Points:

- Model fails to learn important features
- Performs poorly on both training and testing data
- High Bias, Low Variance

Causes:

- Simple model
- Fewer features
- Insufficient training

Effects:

- High training error
- High testing error

Solution:

- Increase model complexity
- Add relevant features
- Reduce regularization

2. Overfitting

Definition:

Overfitting occurs when a model learns both patterns and noise from training data.

Key Points:

- Model memorizes data
- Performs well on training but poorly on test data
- Low Bias, High Variance

Causes:

- Complex model
- Small dataset

- Noise in data

Effects:

- Very low training error
- High testing error

Solution:

- Use more data
- Apply regularization (L1/L2)
- Use cross-validation
- Reduce model complexity

3. Differences

Aspect	Underfitting	Overfitting
Model	Too simple	Too complex
Learning	Insufficient	Excessive
Training Error	High	Low
Testing Error	High	High
Bias	High	Low
Variance	Low	High

4. Bias-Variance Relation

- Underfitting → High Bias, Low Variance
- Overfitting → Low Bias, High Variance

5. Generalization

Definition:

The ability of a model to perform well on unseen data.